Analysis pipeline: what runs, and why

Every branch in one swing's analysis, as a UML activity diagram. Generated by tools/analysis/pipeline_activity.py and checked against the code.

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8 Scoring and write-back — The last stages, then what reaches the document.

Reading a page

computes · loads from the document · started, then refused · a step inside a stage

red diamond: a decision the document does not keep. A black one is kept, or partly kept.

bar: fork or join. A join means all of it is finished before what is drawn next; the executor may overlap more, never less.

Of 19 decisions, 0 leave no record in the document and 2 are only partly kept. The fine print in a box lists what it reads (in) and writes (out), from the executor's export.

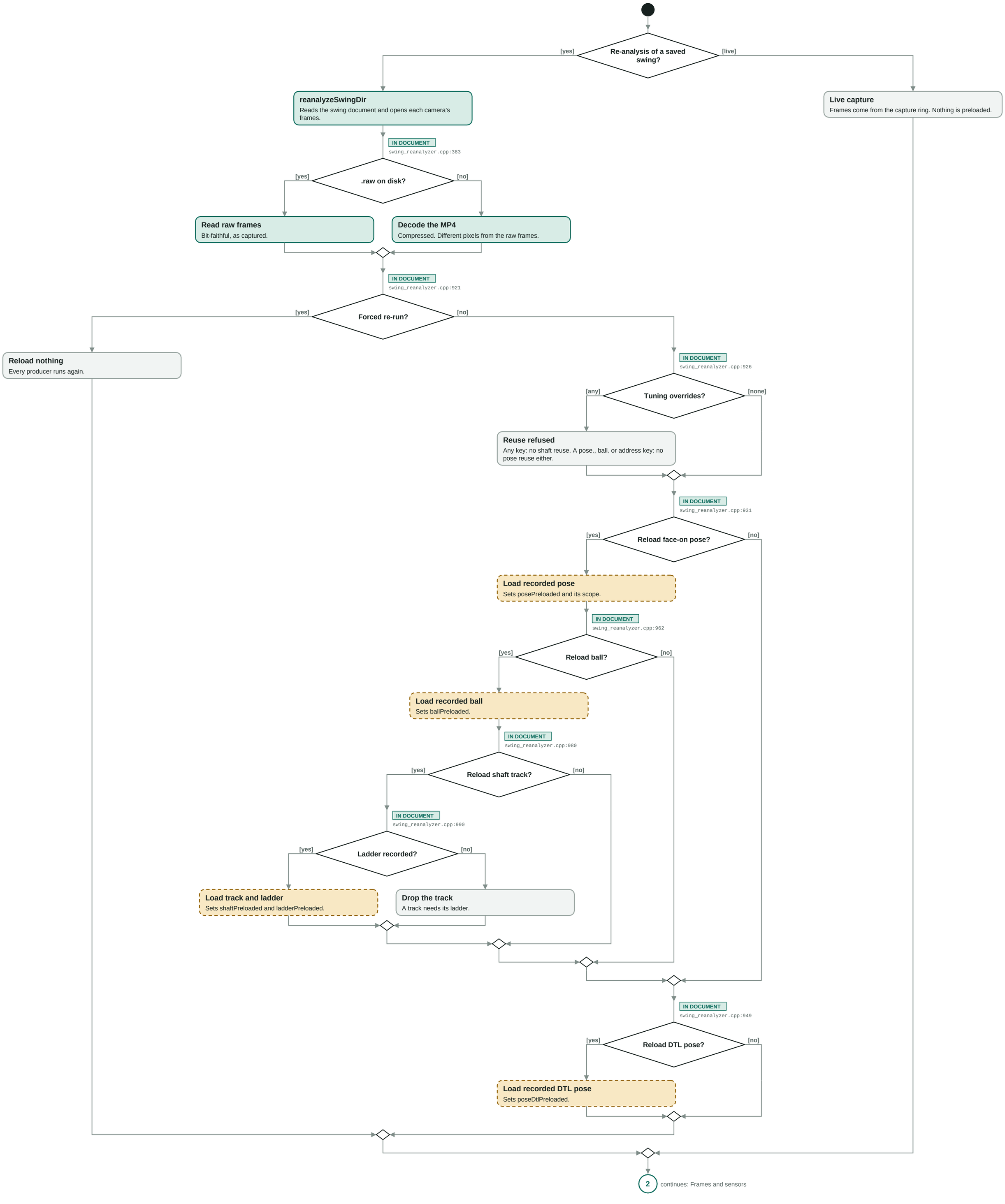
The decisions

Decision	Kept?	Where the answer goes	Why it matters
Implausible samples? `analysis_pipeline.cpp:869`	PARTLY	Partly: demoted samples carry a flag, not that a reuse pass set it.	The checks run again on every reuse, and can demote one more sample than the fresh run did.
Ball found at address? `shaft_tracker.cpp:272`	PARTLY	Partly: club.addressBall is absent when no ball anchored.	With no ball, a fresh run keeps a synth built before P1's angle was rewritten; a reuse rebuilds it.
Forced re-run? `swing_reanalyzer.cpp:921`	IN DOCUMENT	In the document: path.forced, and every reuse verdict says "forced".	swinglab only; the app never sets it. Throws away every recorded track.
Tuning overrides? `swing_reanalyzer.cpp:926`	IN DOCUMENT	In the document: path.overrides lists the keys.	A pose., ball. or address key re-runs pose. Any key at all re-runs the shaft.
.raw on disk? `swing_reanalyzer.cpp:383`	IN DOCUMENT	In the document: path.frames, one answer per camera.	Raw frames or the compressed MP4. Everything downstream sees different pixels.
Reload face-on pose? `swing_reanalyzer.cpp:931`	IN DOCUMENT	In the document: path.reuse.pose, with the first reason if refused. The stage records source: loaded.	Pose version, model file and scope all match, and no pose override.
Reload ball? `swing_reanalyzer.cpp:962`	IN DOCUMENT	In the document: path.reuse.ball, and source: loaded.	Needs a reloaded pose and a matching ball version.
Reload shaft track? `swing_reanalyzer.cpp:980`	IN DOCUMENT	In the document: path.reuse.shaft, and source: loaded.	Reloaded ball, matching shaft version, a recorded track, and NO tuning override of any kind.
Ladder recorded? `swing_reanalyzer.cpp:990`	IN DOCUMENT	In the document: path.reuse.ladder, and each stood-down stage gives its reason.	A track without its ladder cancels shaft reuse. With one, the ladder stages stand down.
Reload DTL pose? `swing_reanalyzer.cpp:949`	IN DOCUMENT	In the document: path.reuse.poseDtl, and source: loaded.	Independent of the face-on decision, so a swing can hold a new face-on pose beside an old DTL one.
Which pose path? `analysis_pipeline.cpp:616`	IN DOCUMENT	In the document: path.pose is bounded, twoPass, twoPassWindow, window or recorded.	The branch #154 was hiding behind.
Coarse pass found a span? `pose_runner.cpp:776`	IN DOCUMENT	In the document: path.pose reads twoPassWindow when no span was found.	If not, pass 2 fills the whole window.
Crop accepted? `pose_runner.cpp:645`	IN DOCUMENT	In the document: pose2d.cropRect, absent when full-frame.	A crop of 90 % of the frame or more, or too few bbox frames, falls back to full-frame.
Tracker span-bounded? `shaft_tracker.cpp:84`	IN DOCUMENT	In the document: path.trackerSpan is bounded, window or recorded.	--full-window turns it off. Series lengths change; positions don't.
DTL witness source? `analysis_pipeline.cpp:888`	IN DOCUMENT	In the document: path.witness is traced or flags.	Shifted 2 of 410 published DTL frames on one swing.
Which ladder wins? `seg_resolve.h:55`	IN DOCUMENT	In the document: path.ladder is recorded, sensors, vision or sensorClamp.	Reused, then adopted sensors, then vision, then the sensor clamp.
refine.fusion on? `analysis_pipeline.cpp:1237`	IN DOCUMENT	In the document: segmentation.fusion.	Picks one of the two ladder builders. On by default.
DTL camera present? `analysis_pipeline.cpp:2256`	IN DOCUMENT	In the document: the streams list, and stages show ran: false.	Without it the DTL branch stands down, and Skeleton3D starts and refuses.
Motion sensors present? `analysis_pipeline.cpp:499`	IN DOCUMENT	In the document: the streams list.	Decides the pose path and which ladder wins.

1 How the run starts

Before any stage runs: what can be reloaded from the swing document. Everything later reads these answers.

Contents

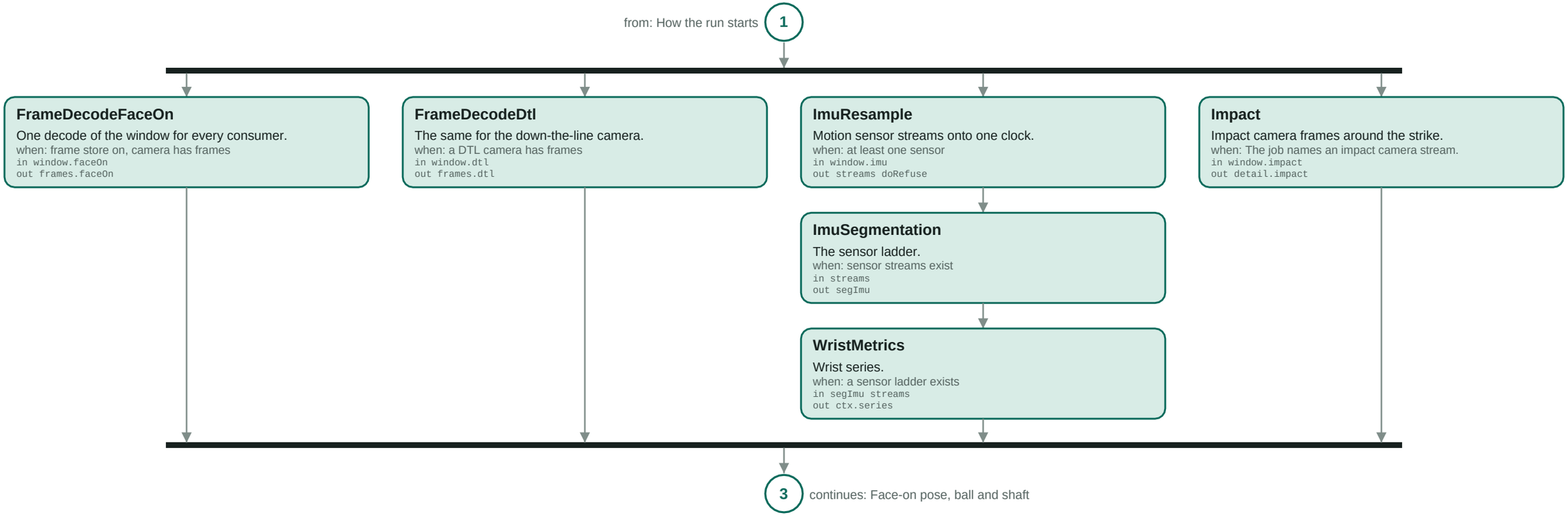


2 continues: Frames and sensors

2 Frames and sensors

Decoded once per camera. The sensors decide the pose path on the next page.

Contents



3 Face-on pose, ball and shaft

The heart of #154: each producer either loads or computes. Since the path record, the document says which.

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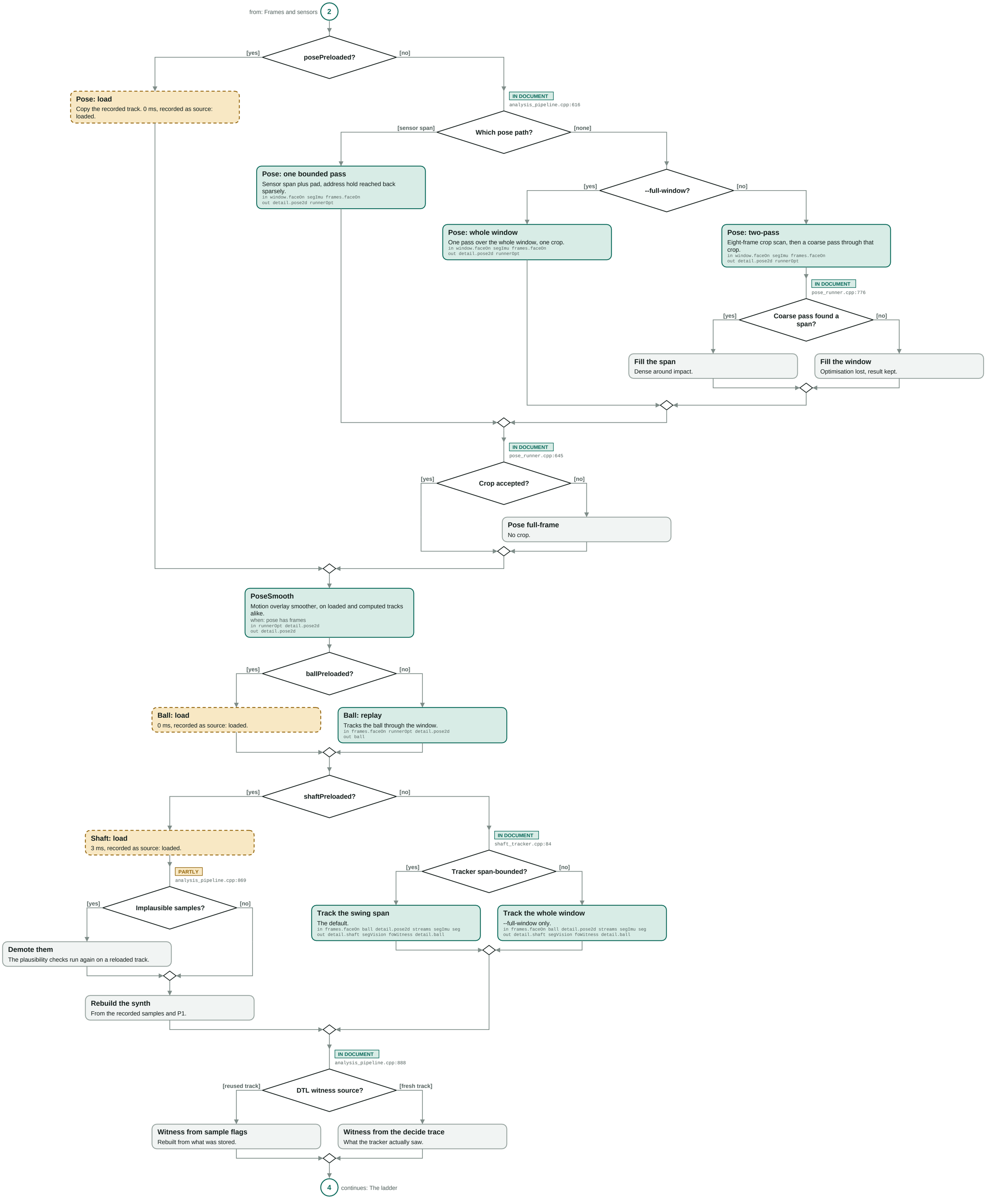
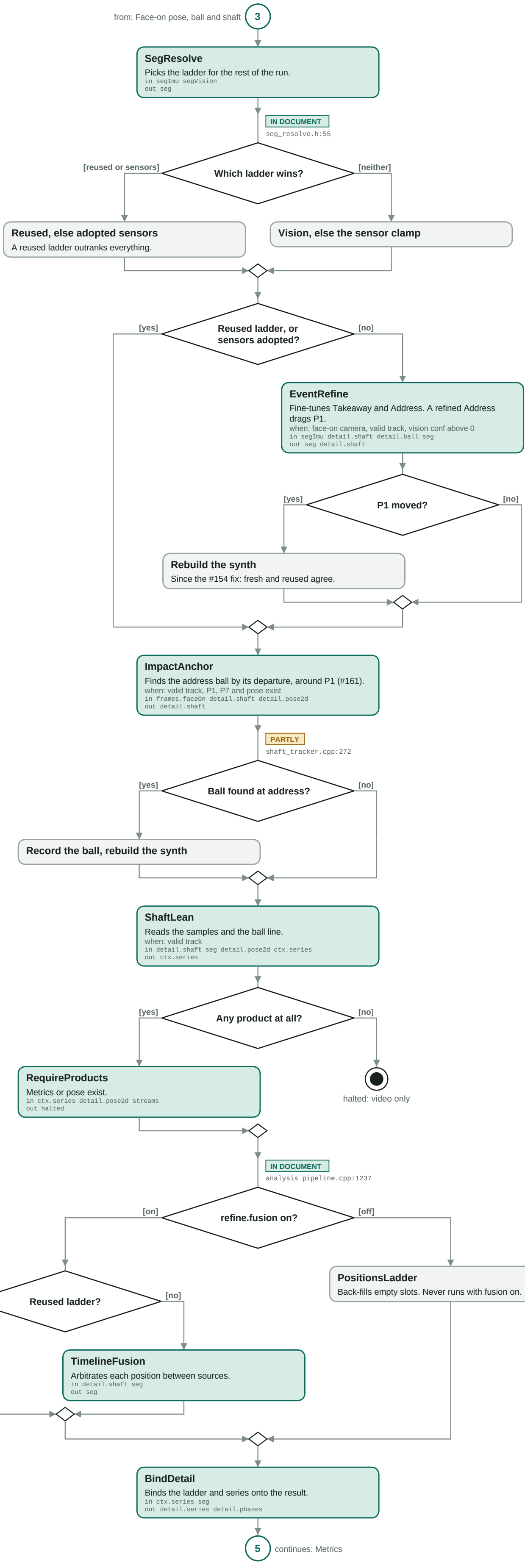


4 The ladder

Contents

One ladder of phases and positions. A reused ladder stands the refiners down.



5 Metrics

Contents

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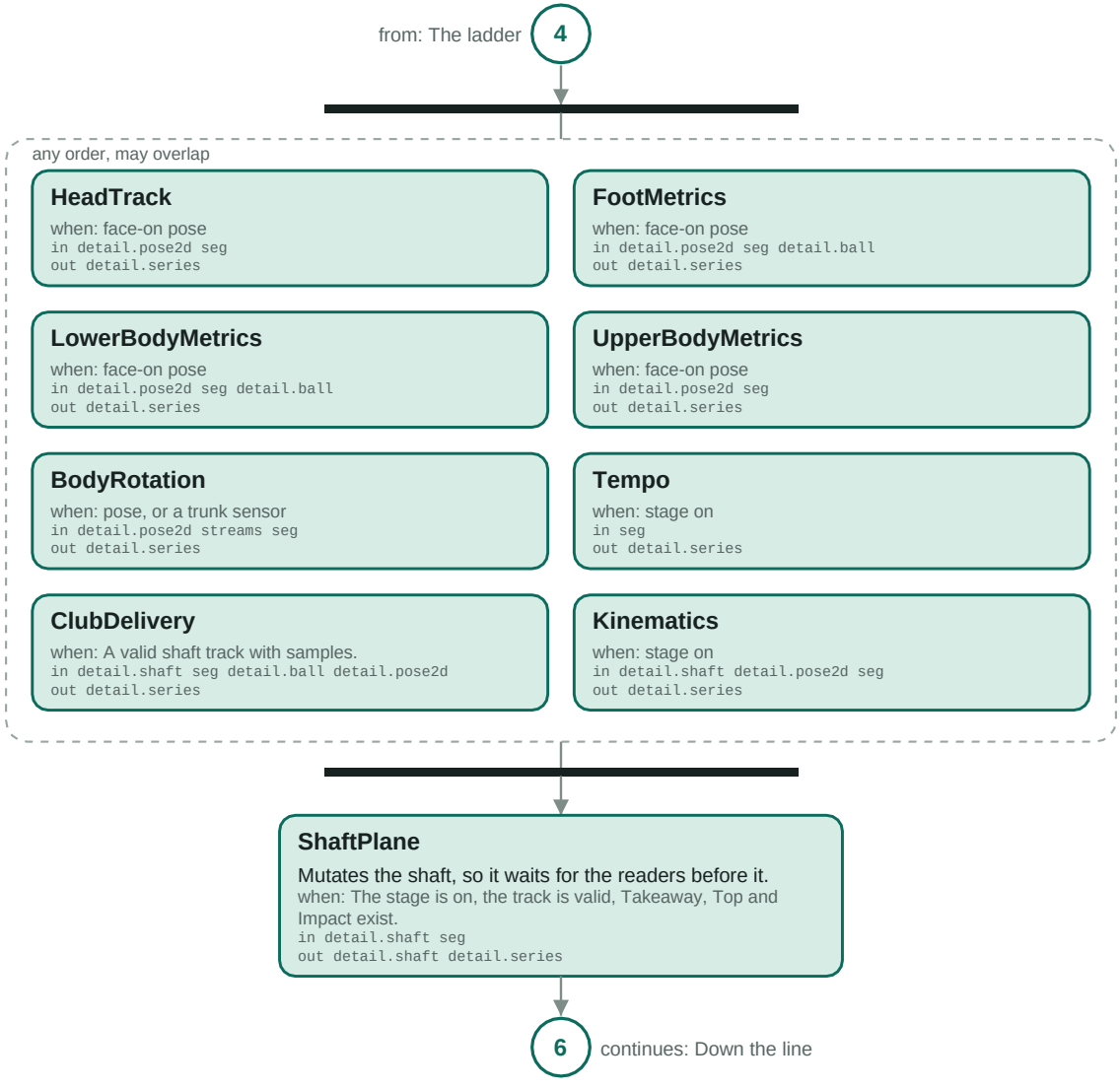
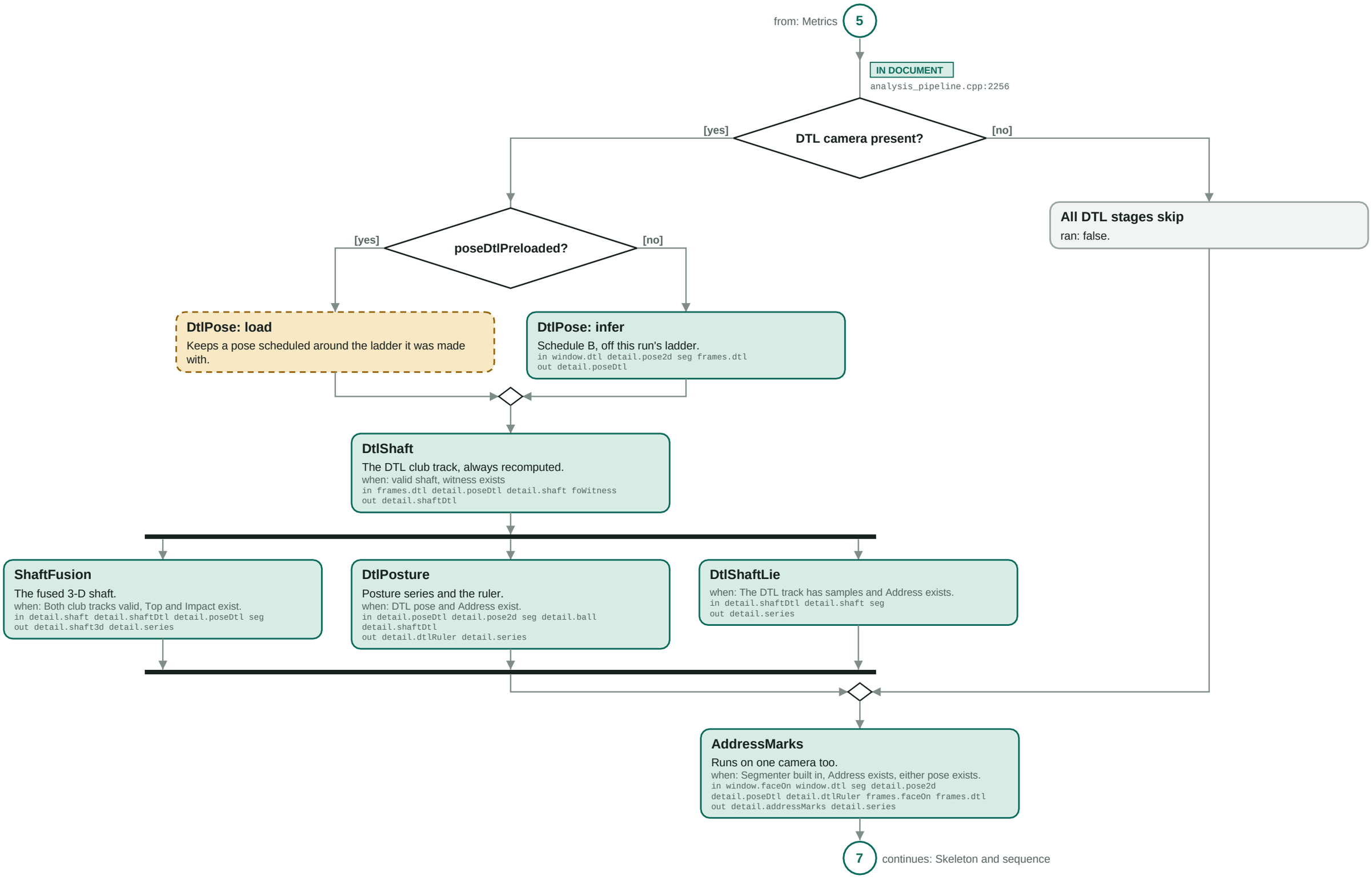


6 Down the line

Everything that needs the second camera. Its pose decision is separate from the face-on one.

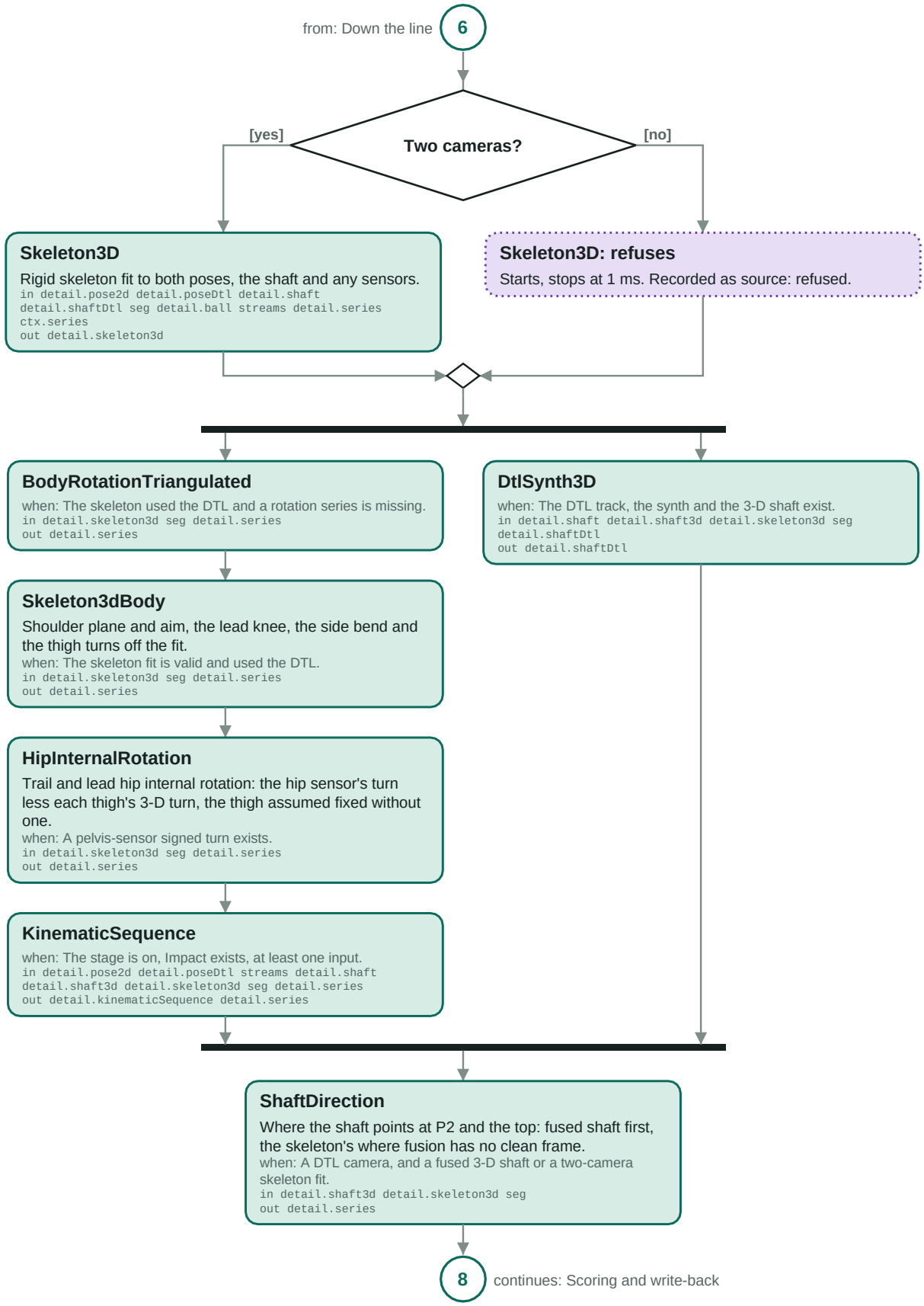
Contents



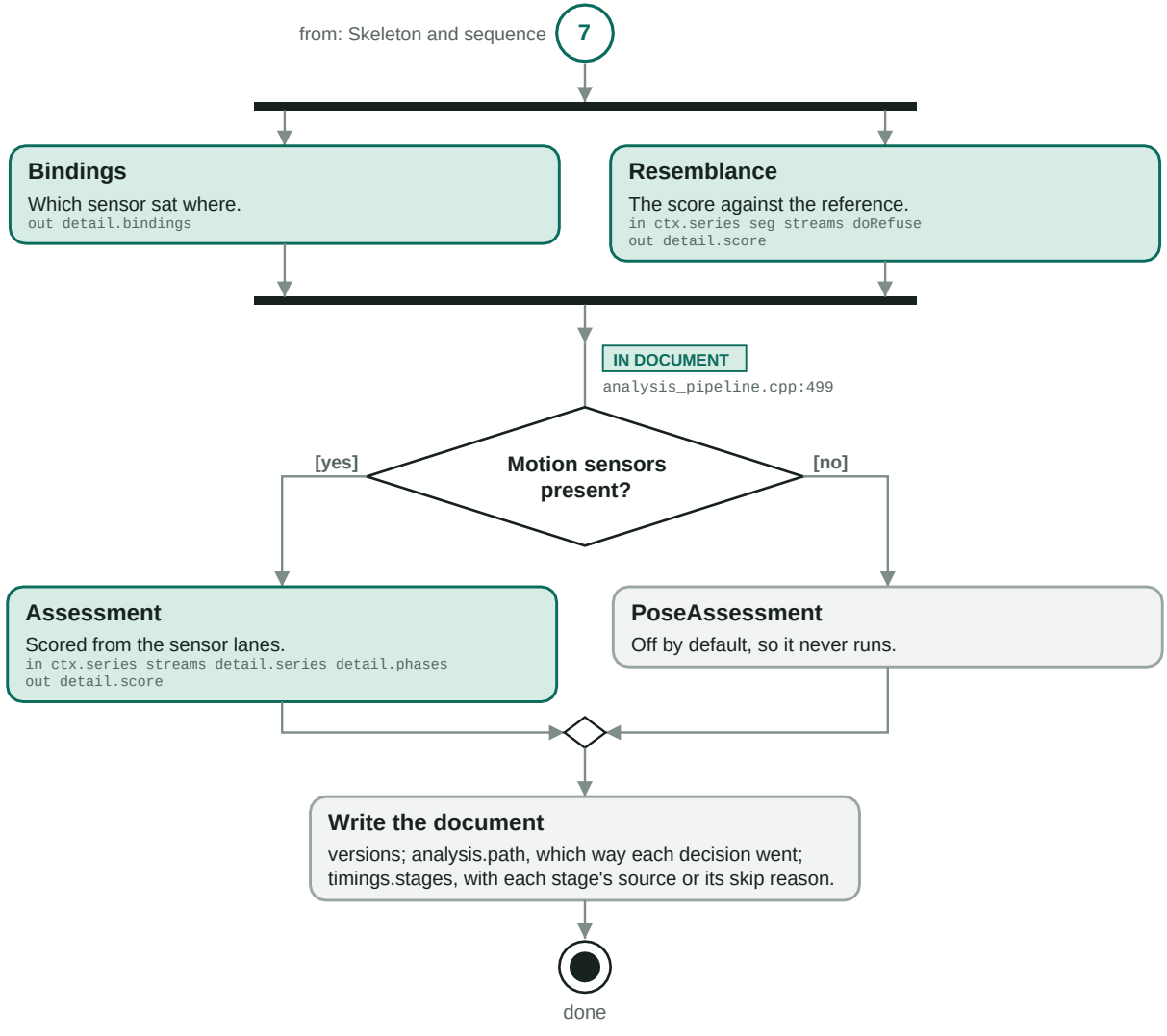
7 Skeleton and sequence

Contents

The 3-D fit needs both cameras, but it starts either way.



8 Scoring and write-back

The last stages, then what reaches the document.

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